

Lab Exercise: Plotly and Dash using the Movie Dataset

Course: Python for Data Science

Topic: Interactive Data Visualization with Plotly and Dash

Dataset: `movies_updated.csv`

Instructions

- Use the provided **movies_updated.csv** dataset.
 - Write clean and well-commented code.
 - Add appropriate chart titles and axis labels.
 - Use Plotly Express to create the visualizations.
 - Save your work in a Jupyter Notebook (`.ipynb`).
 - Build your Dash application in a separate Python file (`app.py`).
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Part I: Plotly

Exercise 1

Load the movie dataset and perform the following tasks:

- Display the first five rows.
 - Display the number of rows and columns.
 - Display the column names.
 - Display the data types of each column.
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Exercise 2

Create a **bar chart** showing the number of movies in each genre.

Requirements:

- Use different colors for each genre.
 - Add a chart title.
 - Label the x-axis and y-axis.
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Exercise 3

Calculate the average movie score for each genre and create a **bar chart**.

Requirements:

- Color the bars using the average score.
 - Add a suitable chart title.
 - Display genres on the x-axis.
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Exercise 4

Create a **scatter plot** showing the relationship between movie budget and gross revenue.

Requirements:

- Color the points by genre.
 - Display the movie name when hovering over a point.
 - Add an appropriate chart title.
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Exercise 5

Create a **line chart** showing the number of movies released each year.

Requirements:

- Use markers.
 - Add axis labels.
 - Add a chart title.
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Exercise 6

Create a **histogram** showing the distribution of movie scores.

Requirements:

- Choose an appropriate number of bins.
 - Add a chart title.
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Exercise 7

Create a **pie chart** showing the distribution of movie ratings (PG, R, PG-13, etc.).

Exercise 8

Find the **Top 10 Highest Grossing Movies** and visualize the result using a **horizontal bar chart**.

Requirements:

- Sort movies by gross revenue.
 - Display movie names on the y-axis.
 - Color the bars based on gross revenue.
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Part II: Dash

Exercise 9

Create a simple Dash application containing:

- A page title
- One Dropdown for selecting a movie genre
- One Plotly graph

When the user selects a genre, update the graph to display the average movie score by rating.

Exercise 10

Extend the previous application by adding another filter.

Choose one of the following:

- Country
- Rating
- Director

The graph should update whenever either filter changes.

Exercise 11

Add a second graph to the dashboard.

Choose one of the following:

- Budget vs Gross Revenue
- Movies Released Per Year
- Genre Distribution
- Movie Rating Distribution

Both graphs should respond to the selected filters.

Exercise 12

Add a **Year Range Slider** to the dashboard.

Requirements:

- Allow users to select a range of years.
 - Update all graphs automatically based on the selected year range.
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Exercise 13

Improve the appearance of your dashboard by adding at least three of the following:

- Better chart titles

- Different chart colors
 - Hover information
 - Horizontal bar charts
 - Pie charts
 - Dashboard styling using `html.Div`
 - Multiple sections or columns
 - Appropriate spacing between components
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Final Challenge

Build a **Movie Analytics Dashboard** using Dash.

Your dashboard should include the following filters:

- Genre
- Country
- Rating
- Year Range

Include at least **six** interactive visualizations, such as:

- Movies by Genre
- Average Score by Genre
- Movies Released Per Year
- Budget vs Gross Revenue
- Rating Distribution
- Top 10 Highest Grossing Movies
- Average Runtime by Genre
- Top Directors by Number of Movies

All visualizations should update automatically whenever the user changes any filter.